



Telehealth

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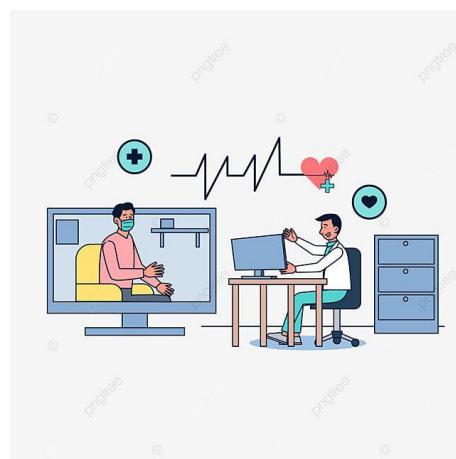
Introduction

Since their existence, humans have been trying to improve, grow, and develop all aspects of life. One of the major aspects that humans have worked on to advance is health, and we can see the effort that the past generation put into developing health in today's world. The new tools, books, methods, and the thing that flipped the major upside down is the engagement of technology in the healthcare field.

Telehealth is one of the ways to provide healthcare that is technology-enabled. It opened a huge door of services and aids to help patients get high-quality care without even being physically in the hospital.

Telehealth can be defined as the utilization of communication technologies and official platforms to give health care and attention to anyone in need but from a distance. It could be in the form of live videos where a patient and a physician enter a private room and discuss the patient's issue, or in the form of educational images, videos, or online published articles. In addition to the use of Remote Patient Monitoring (RPM) which are devices used to detect data about a patient's health state such as vital signs.

The invention of Telehealth has helped the health system a lot, and it has several implementations in the health care sector, but because we can't always be certain that technology only has a positive side and effect, Telehealth still has some disadvantages that need to be taken into consideration.



Meaning, History, Platforms of Telehealth.

The history of telehealth started in 1948 when the first radiological images were sent to other specialists. In 1959, it was the first record of sending health documents forth and back between healthcare workers across the country by the University of Nebraska. In 1991, Dr. Jay Sanders, who is considered the father of telehealth developed the first telehealth system in the state of Georgia. In addition, he created the first telemedicine program named " The Electronic House Call".Dr. Jay Sanders was the Founder of the American Telemedicine Association. Dr. Rashid Bashshur arranged the first conferences on Telemedicine and published them in 1975 and he became the president of the American Telemedicine Association for two years. After the Internet network was born in 1983. The American Telemedicine Association was founded in 1993, which is an organization that promotes and improves telehealth to allow the reach of health to a large number of patients and those who have difficulty and struggle to access it.



Telehealth platform is used to gather, process, and analyze health data collected from measuring devices and the interview between healthcare provider and patient. It helps the healthcare provider to monitor the patient's health state remotely. Telehealth platforms allow the patient to get online appointments and enter a private room with the doctor for consultation. All telehealth platforms have strict standards and guidelines to protect patients' healthcare information to ensure patient autonomy and confidentiality. Telehealth platforms allow the patient to see all her/his medical records such as medicine, lab results, and radiological results. The essential three components of telehealth are data management systems, monitoring devices, and communication technologies. They form a

framework to support secure data management, improve the healthcare delivery to the patient, and ensure effective communication. Improving the telehealth platform through integrating augmented reality (AR) and virtual reality (VR) to simulate and enhance the interactions between patients and healthcare providers. Many telehealth platforms have different features and have a goal to allow high-quality healthcare delivery. For example, Mend is a popular telehealth platform that provides a lot of services for its clients such as a virtual waiting room, customizable branding, and integration with electronic health records. The second platform is Doxy which is an excellent software that provides simple and secure telehealth without the need to download the software and has many features such as HIPAA-compliant software. Finally, AMC Health provides virtual care coordination and is designed to reduce the cost of healthcare and improve the quality of healthcare delivery for the patients.

Implementation of Telehealth in Healthcare.

Telehealth implementations in healthcare practice have created a major change in medical service delivery by enhancing accessibility to care while improving operational efficiency and patient treatment quality. Remote healthcare service delivery relies on digital communication technologies combined with video conferencing systems and mobile health applications and remote monitoring devices together with electronic health records. This modern medical method has proven crucial for delivering services to patients who lack proper healthcare access in distant areas. Telehealth enables patients to receive healthcare professional consultations from their location through virtual visits and thereby removes geographical

limitations thus guaranteeing fast medical help essential for early illness detection and disease control as well as primary medical care.

One of the primary benefits of Telehealth increases patient outcomes through automated healthcare monitoring services combined with immediate medical care. The use of wearable devices by chronic illness patients enables continuous monitoring of vital signs which allows automatic health data transmission to physicians. Through telehealth healthcare providers gain early access to detect anomalies adjusting possible which decreases hospitalization rates alongside emergency room visits. Since telehealth helps patients access mental health assistance through online therapy, they can receive psychological support without encountering the difficulties that come with seeking medical help in person. Remote consultation methods from physicians serve elderly patients and patients with mobility challenges well by providing medical support directly to their home environment.

Another major advantage of Telehealth programs enables hospitals to achieve significant cost reduction through reduced practice operations across both provider funds and patient finances. Telehealth reduces medical facility load because it minimizes inappropriate hospital trips which enables healthcare providers to use resources for optimal effect. Patients improve their financial situation because they reduce expenses related to travel and they eliminate both the long-term waiting periods and work-related cost from their medical care. The management of public health emergencies becomes possible through telehealth services as proven by the COVID-19 pandemic. Healthcare systems maintained their operations through fast telemedicine implementation that lowered the chance of virus spread.

Virtual consultations emerged as the preferred form of medical service which boosted both healthcare provider and patient acceptance of telehealth services. The widespread adoption

of telehealth requires resolution of multiple obstacles before reaching global implementation. Residents who reside in low-income communities' face challenges by accessing reliable internet services and possessing digital devices along with the required technological abilities. Patient information security stands as a major concern during the process of sharing healthcare data online. Each health care location maintains separate rules and insurance regulations regarding telehealth which creates barriers to its consistent implementation. The resolution of these issues depends on improved internet access together with defined regulations and educational programs about telehealth utilization.

Advantages of Telehealth

The use of Telehealth has enabled healthcare providers to reach a wide range of patients regardless of their situation whether they have mobility limitations such as elderly, people with injuries, people on wheelchairs or even paralyzed people. In addition to the patients that live in remote areas where there are no clinics or a health care provider that can give sufficient medical care. Also, population that are under occupation and war where healthcare settings are being destroyed, and healthcare providers are being killed.

Telehealth can decrease the opportunity of contagious diseases spreading. Because both health care provider and the patients will be in completely different areas and maybe different continents and there will be no chance at all for a disease to be transmitted not by direct touch, touch of contaminated surface, droplets or any other way. Also, it will protect the people who are immunocompromised by decreasing the amount of outpatient visits as any germ, virus, bacteria may cause them a serious problem (Watson, 2020).

Telehealth can be one of the ways to reduce the time and the money that are being wasted during transportation to and from the hospital. In addition to the waiting time in the hospital for your turn. Moreover, it will decrease the usage of medical services and devices in healthcare settings as this process can be replaced by the Remote Patient Monitoring programs specifically for patients' chronic diseases such as diabetes and hypertension.

Disadvantages

Although telehealth has many advantages, it also has drawbacks and weaknesses that need improvement, including:

First, doctors and nurses cannot physically examine patients or perform hands-on assessments to test them, which can lead to misdiagnosis. Also, some diseases primarily require specific tests and close examinations, such as palpation and percussion, or listening to sounds such as heartbeats or lung sounds. Therefore, telehealth may not be a sufficient solution.

Second, technology can be a barrier for older patients who may not know how to use it and have no one to help them. Patients with limited internet access or living in areas without internet access may be unable to receive the necessary care. Without the internet, there is no telehealth.

In addition, telehealth is vulnerable to security breaches because it relies on digital platforms. Therefore, patients may be concerned about the hacking and misuse of their personal information and medical records. Another major drawback is the difficulty of handling emergency cases. In cases such as a heart attack or stroke, healthcare cannot be provided remotely immediately. Doctors may face significant difficulty and an inability to aid remotely.

Finally, the drawbacks of telehealth include not only misdiagnosis due to the inability to perform a clinical examination but also misunderstanding of medical instructions and improper healthcare delivery by the patient or their supervisor. For example, suppose a patient suffers from diabetes and, based on the information they provided to the doctor remotely, the doctor prescribes a dose of insulin. However, the doctor or their family may have miscalculated, or the patient may have been given the wrong dose or injection, which could lead to a dangerous rise or fall in blood sugar that may require hospitalization and potentially put the patient's life at risk.

Compare Between Telehealth and Direct Health Settings

Face-to-face health centers and telehealth are two different modes of delivery of healthcare services.

A comparison between the two is given below:

-Access to care:

Telehealth has revolutionized access to care by opening an online avenue through which patients can access medical care without having to overcome conventional barriers of distance, mobility, and shortage of healthcare facilities. Through the use of digital platforms such as video calling, telephone consultations, and secure messaging, telehealth is taking healthcare to more people with convenience, proximity, and ease.

Direct health setting access is where the patients personally go to health facilities like hospitals, clinics, or physicians' offices to get healthcare treatment. It forms the face-to-face care, consultation, and diagnostic tests.

-Quality of Care:

Telehealth: The quality of telehealth care is the effectiveness, safety, patient satisfaction, and outcomes of care provided by remote digital technology. For all the extent to which telehealth increases access and convenience, high standards of quality of care are fundamental to its success as a model of healthcare delivery.

Direct Health Setting: Patient-centered care in the direct health setting is described as giving first consideration to patients' values, needs, and preferences in healthcare. It is a way of making the patient a full partner in decision-making regarding his/her own healthcare, its effectiveness, safety, and satisfaction with the received healthcare.

-Cost:

Telehealth: Cost-effectiveness in telehealth is achieved through the delivery of healthcare services over a distance in an attempt to reduce the cost of healthcare for both patients and healthcare providers. It is achieved by saving on transport costs, more efficient utilization of healthcare resources.

Direct Health Setting: Overhead in a health care facility involves continual outlay in the form of manpower, building upkeep, medical technology, and other operational requirements. They are part of face-to-face health care service cost and have a profound effect on overall cost to individuals and the health care system.

-Technology Requirements:

Telehealth: relies on electronic technology to provide remote health care. Both the patients and the healthcare professionals need to be endowed with the requisite technologies to

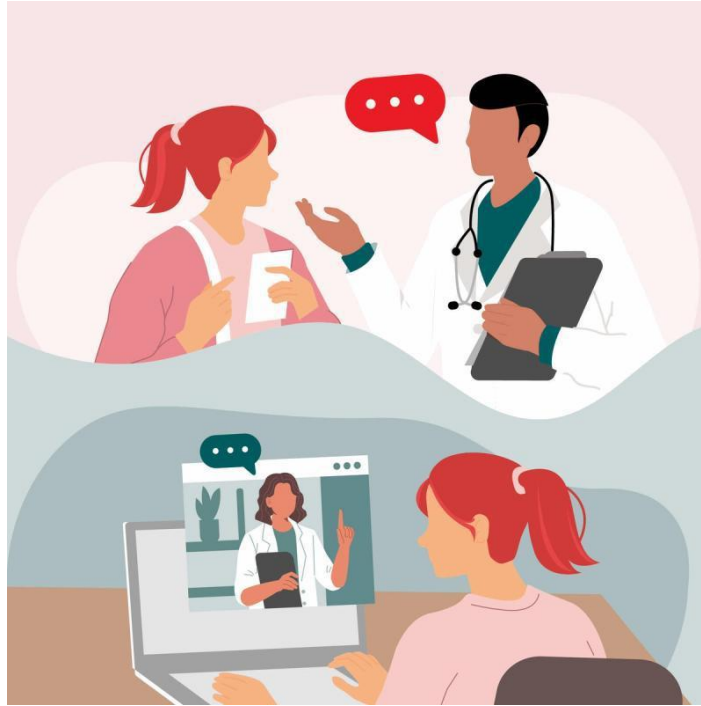
allow effortless communications, secure information, and offer health care effectively for telehealth to succeed.

Direct Health setting: Technology in the direct health setting, i.e., hospitals, physician practices, and clinics, is at the forefront of quality-of-care improvement, patient safety, and efficiency. It underpins the main functions of healthcare delivery, viz. treatment, diagnosis, and patient monitoring.

-Limitations:

Telehealth: There are also some patients who will not be able to use telehealth as they lack the requisite devices (computers, smartphones, or internet-enabled devices). Poor quality internet access and low speeds, especially among impoverished and rural communities, will also hinder their effective use of telehealth services.

Direct Health Setting: Geographical restrictions can be a challenge to those residing in rural or remote areas in the context that direct healthcare is hard to access owing to distances, and transportation challenges.



Conclusion

In conclusion, there is no doubt that telehealth is a valuable step in healthcare, providing significant assistance during crises, pandemics, and when there is a potential for the transmission of serious infections. However, it needs to be developed and overcome challenges in cases requiring rapid intervention or direct clinical examinations. However, this does not prevent its use in many other areas, including improving it and enhancing its cybersecurity. While it is not a complete alternative to direct treatment, it may be a lifeline in cases where access to traditional treatment is difficult. Therefore, it is best to strike a balance between the two.

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